

JohnnySonic
Score File Cheat Sheet

!Comment	Any line starting with a "!" is ignored. Comments can appear anywhere in the score file
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	min	max	default	
--	-----	-----	---------	--

/Chorus	Invoke the choruser on a segment of the sound buffer			
startBeat	0			starting beat
duration	>0			duration in beats
numberOfVoices				typically an even number (voices are divided across channels)
depth		<1		comb filter gain
flipChannels			0	1=chorussed right chan is output to left, and vice-versa

Example:

```
/Chorus
!st  dur  voices  depth  flipChan
8.0  12.5  4      0.9    0
```

Notes: This is a basic FIR choruser using comb filters with gradually changing lengths

	min	max	default	
--	-----	-----	---------	--

/Clip	Initiate clip mode			
clipChanNumber	0	1		0=left; 1=right
clipName				string. Must match the clip name in the clipList file

Example:

```
/Clip
!chan  clip
1      ExtSound
!st  dur  vol  loc  clStart  clRate  Inst
5.1  2.2  5   0.5  0.1      1.1    ClipTest
```

Notes: Clip mode changes the meaning of the 5th and 6th parameters for notes

	min	max	default	
--	-----	-----	---------	--

/Compress	Invoke the compressor on a segment of the sound buffer			
startBeat	0			starting beat
duration	>0			duration in beats
maxRatio	2			maximum compression ratio

Example:

```
/Compress
!st  dur  maxRatio
8.8  12.0  5
```

Notes: Primarily used to tame large transient peaks
MaxRatio typically in the range of 3 .. 15

/End (End of score file)	This should be the last command in the score file
(no parameters)	

Example:

```
/End
```

Notes: The /End command can also be anywhere in the score file to stop the processing at that point

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/Exclude	This should be the last command in an INCLUDE file
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(no parameters)

Example:

/Exclude

Notes: The /Exclude command can also be placed anywhere in the include file to cause the processing to continue where it left off in the main score file

	min	max	default
--	-----	-----	---------

/FBA (Frequency Band Analyzer)	Generate file of VU data by frequency band
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startBeat	0	starting beat for the display
Duration	>0	duration in beats
channel		0 = Left; 1=Right; -1= Combined
buffer		S s = soundBuffer; M m = mixBuffer

Example:

/FBA

lst	dur	chan	buf
10.0	12.5	0	s

Notes: /ShowBuffer creates an output file. It makes no changes to the sound buffer
 Dump file name is hardcoded as Data/FBADump
 Dump file contains RMS by frequency band in 0.25 second buckets

	min	max	default
--	-----	-----	---------

/Filter	Apply an audio filter on a segment of the sound buffer
----------------	--

startBeat	0	starting beat (-1 to continue where previous line left off)
duration	>0	duration in beats
filterType	1 16	See below for filter types
initialPitch		positive = pitch number; negative = frequency in Hertz
finalPitch		positive = pitch number; negative = frequency in Hertz
initialBandwidth		positive = number of semitones; negative = frequency in Hertz
finalBandwidth		positive = number of semitones; negative = frequency in Hertz

Example:

/Filter

lst	dur	type	iPitch	ePitch	iBW	eBW
8.0	20.0	5	-550	-550	-80	-80
8.0	20.0	5	-550	-550	-40	-40
30	20.0	4	-400	-400	-1	-1
30	20.0	4	-700	-700	-1	-1
55	20.0	14	-500	-500	-1	-1

Notes: Bandwidth parameters apply only to filter types 1, 2, 5, 6

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Filter Types:

1	all-pole bandpass with zero dB gain at passband
2	all-pole bandpass with power output equal to power input
3	Butterworth lowpass
4	Butterworth highpass
5	Butterworth bandpass
6	Butterworth notch
13	FIR lowpass
14	FIR highpass
15	Not implemented yet
16	FIR notch (LP @ .91 x freq + hp @ 1.1 x freq)

	min	max	default
/Flange Invoke the flanger on a segment of the sound buffer			
startBeat	0		starting beat
duration	>0		duration in beats
numberOfVoices			typically an even number (voices are divided across channels)
depth		<1	comb filter gain
flipChannels		0	1=flanged right chan is output to left, and vice-versa

Example:

```

/Flange
!st dur voices depth flipChan
8.0 12.5 4 0.9 0

```

Notes: This is a basic IIR flanger using comb filters with gradually changing lengths

	min	max	default
/GEQ (Graphic Equalization) Invoke the graphic equalizer on a segment of the sound buffer			
beat	0		starting beat
band1Gain			positive or negative in dB
band2Gain			positive or negative in dB
...			
band30Gain			positive or negative in dB

Example:

```

/GEQ
!st G1 G2 G3 G4 G5 G6 G7 G8 G9 G10 G11 G12 G13 G14 G15 ... G30
15.0 3 2 0 -5 -5 -2 0 0 2 3 3 1 0 0 0 -2
37.0 3 2 0 -5 -5 -2 0 0 2 3 3 1 0 0 0 -2

```

Notes: Only partially implemented

Takes two (or more) lines of parameters

First line determines starting beat and the gains for all 30 bands

Second (or final) line determines ending beat

To be implemented: interpolating the gains

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min max default

/Haas Set global Haas setting

HaasEnabled	0	1		1=Haas enabled; 0=Haas not enabled
minDelay	0	40	0	minimum delay in mS
maxDelay	0	90	90	maximum delay in mS
reverbSend	0	1		1=true; 0=false

Example:

/Haas

!enabled min max reverbSend

1 20 50 0

Notes: This command effects all subsequent notes until changed by another /Haas command

Haas is disabled by default at the start of the program

/Include Process an include file before contuing with the main file

incFileName

Example:

/Include

Data/Ivor2

Notes: A score file can contain any number of include files

Include files cannot be nested

min max default

/Levels Adjust levels on a segment of the sound buffer

startBeat	0			starting beat (-1 to continue where previous line left off)
duration	>0			duration in beats
startGainLossdB				positive or negative adjustment in dB
endGainLossdB				positive or negative adjustment in dB

Example:

/Levels

!st dur stGain/LossIndB endGain/LossIndB

8.8 2.0 0 -10

-1 3.0 -10 0

Notes: This uses linear interpolation (not ideal)

min max default

/Mix Mix Sound and Haas buffers to the Mix buffer

startBeat	0			starting beat (-1 to continue where previous line left off)
duration	>0			duration in beats
gainLossdB				positive or negative adjustment in dB
keepSoundFlag			0	1=keep sound/haas buffers; 0=clear them
sign			1	use -1 to invert the sign
timeOffset			0	in seconds (not beats). Can be positive or negative

Example:

/Mix

!st dur Gain/LossIndB keepFlag sign timeOffset

8.0 20.0 -12 0 1 0

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Notes: There must be at least one /Mix command in the score file

Note to JGP: timeOffset was not yet implemented in the GitHub version

	min	max	default	
				/Pitches Restore default (pitch) mode -- turn off clip, vocode
startBeat	0			starting beat
duration	>0			duration in beats
volume			1	absolute (not dB)
location	-1	1	0	
startPitch				pos = pitch number; neg=frequency in Hertz
endPitch				pos = pitch number; neg=frequency in Hertz
Instrument				string. Must match name in Instrument file
Example:				
/Pitches				
!st	dur	vol	loc	sPit ePit Inst
5.1	2.2	5	0.5	47 47 FiltNz0
5.2	2.1	3.4	-0.8	50 50 FiltNz0

	min	max	default	
				/Pulse Insert a one-sample pulse into the sound buffer
startBeat	0			starting beat
channel				L l = left; R r = right; B b = both
Example:				
/Pulse				
!st	chan			
10.0	L			

Notes: A pulse is useful for testing processors (such as reverb, filters)

	min	max	default	
				/Reverb Invoke the reverberator on a segment of the sound buffer
startBeat	0			starting beat
duration	>0			duration in beats
direction				1=forward (reverb); -1=backward (preverb)
size	1	3		1=small (13 - 23 ms); 2=medium (23 - 40 ms); 3=large (40 - 71 ms)
reverbTime	>0			60 dB reverb time in seconds
combFilterDryGain	0	1	1	gain on dry signal (Shroeder / Moorer used 1.0)
xTalkFactor	0	1		cross-channel gain factor
wetness	0	1		mix of wet/dry on output

Example:

```
/Reverb
!st dur dir size RevTime CFDryGain xTalk wetness
8.8 12.0 1 3 1.2 0.8 0.2 0.6
```

Notes: For very dense or very long reverberation, call /Reverb multiple times

The reverberator is missing the early echos of natural room reverb. I will be enhancing it soon

min max default

/ScreenOut (set screen output level)

screenLevel 0 3 2 0=quiet; 1=medium; 2=verbose; 3=debug

Example:

```
/ScreenOut
1
```

Notes: This overrides the default screenoutput level of 2. It can be changed repeatedly throughout the score

min max default

/SendBack

Send a segment of the mix buffer back to the sound buffer

startBeat 0 starting beat
duration >0 duration in beats
gainLossdB positive or negative adjustment in dB

Example:

```
/SendBack
!st dur Gain/LossIndB
0 20.0 0
```

Notes: Sound buffer is overwritten (at the specified segment)
The specified segment of the mix buffer is cleared

/SFN (Sound File Name)

override the sound file name specified on the command line

soundFileName name of the output sound file to be created in this run

Example:

```
/SFN
Data/MyMasterpiece.AIFF
```

Notes: This allows for a script with the scorefile as the only parameter

min max default

/ShowBuffer

Display the raw sample data for a segment of the sound buffer

startBeat 0 starting beat for the display
Duration >0 duration in beats

Example:

```
/ShowBuffer
!st dur
0 20
```

Notes: /ShowBuffer displays data on the screen. It makes no changes to the sound buffer

min max default

/Stats

Display peak data for a segment of the current sound buffer

startBeat 0 starting beat for the analysis
Duration >0 duration of the analysis

Example:

```
/Stats
!st dur
0 20
```

Notes: /Stats displays information on the screen. It makes no changes to the sound buffer

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	min	max	default
/Tempo	One or more lines defining the tempo (or change in tempo)		
startBeat	0		starting beat for the tempo or change in tempo
Duration	1		duration of the tempo change
initialTempo	>0		initial tempo in beats/minute
finalTempo	>0		final tempo in beats/minute

Example:

```
/Tempo
!st dur init final
0 1 57 57
137 9 57 69
146 2 69 48
148 1 48 57
149 1 57 57
```

Notes: Tempo changes use linear interpolation

If no /Tempo is specified, the default of 60 is used

Limitations: Tempo changes occur on whole-number beats and cannot be instantaneous

currently, songs are limited to no more than 8192 beats. (This can be changed in the program if needed.)

Questions:

which of the parameters can be non-integer?

	min	max	default
/Tuning	One line defining the tuning parameter to be used		
primaryInterval	>1		2
notesPerInterval	>1		12
pitchConvExponent	>0		3
pitchConvFactor	>0		1.021974864

Example:

```
/Tuning
!int notes exp fac
!19-TET
1 19 3 1.021974864
```

Notes: This allows pitches to be specified in any EQUAL-TEMPORED tuning

For non-equal-tempored tunings, pitches should be specified in Hertz (indicated by negative values)

The /Tuning command can be used multiple times in one score file

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min max default

/Vocode

Initiate vocoder mode

clipChanNumber	0	1		0=left; 1=right
clipName				string. Must match the clip name in the clipList file
clipRate			1	positive (forward) or negative (backwards) real number
maxHarm			0	0=limit by nyquist; 1=fundamental only; 2.. =specific limit
slope	-8	8	0	dB / octave slope to be applied to the gain
bassBoost	-15	15	0	in dB. Boost / cut bass frequencies by this amount
dynExponent	0.25	4.0	1	exponent used to expand or compress dynamic range
shiftN	-15	15	0	shift the spectrum up/down N number of bands (out of 16)
peakReduction	0	12	0	In dB. Reduce the loudest formant by this amount

Example:

/Vocode

!chan	clip	clipRate	maxHarm	slope	bassBoost	dynExp	shiftN	peakRed
1	ExtSound	1	0	2.2	4.0	1.1	0	3.0
!st	dur	vol	loc	pitch	clStart	Inst		
5.1	2.2	5	-0.5	-120.0	1.1	VanON		

Notes: Vocoder mode changes the meaning of the 6th parameter for notes (from ending pitch to clipStart)